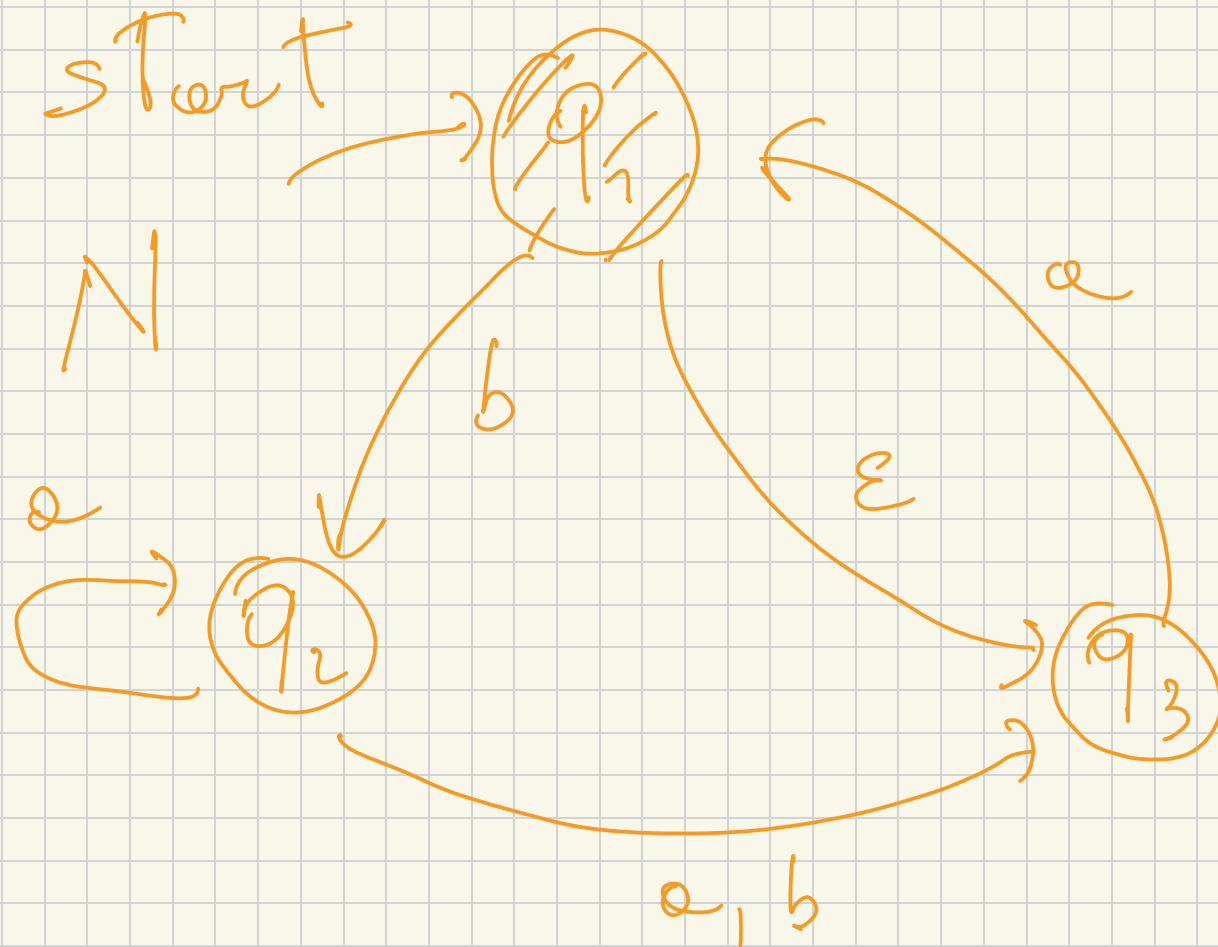


EXERCICIO



$$\Sigma_{\epsilon} = \{a, b, \epsilon\}$$

Transverse DFA

EQV NFA & NFA.

$$M : Q_M = \{q_{\emptyset}, q_{\{1\}}, q_{\{2\}}, q_{\{3\}}\}$$

$$q_{\{1,2\}}, q_{\{1,3\}}, q_{\{2,3\}}, q_{\{1,2,3\}}$$

$$q_0^M = q_{\{1,3\}}$$

$$F_M = \{ q_{\{1\}}, q_{\{1,2\}}, q_{\{1,3\}}, q_{\{1,2,3\}} \}$$

$$S_M : Q_M \times \Sigma \longrightarrow Q_M$$

Considero il diverso caso:

- Nello stato q_2 se N legge a vuoto
trasferisce da $q_2, q_3 : S_M(q_2, \epsilon) =$

$$q_{\{2, 3\}}$$

Somiglianza, $S_{\pi}(q_{\{2\}}, b) = q_{\{3\}}$

- Nello stato q_1 :

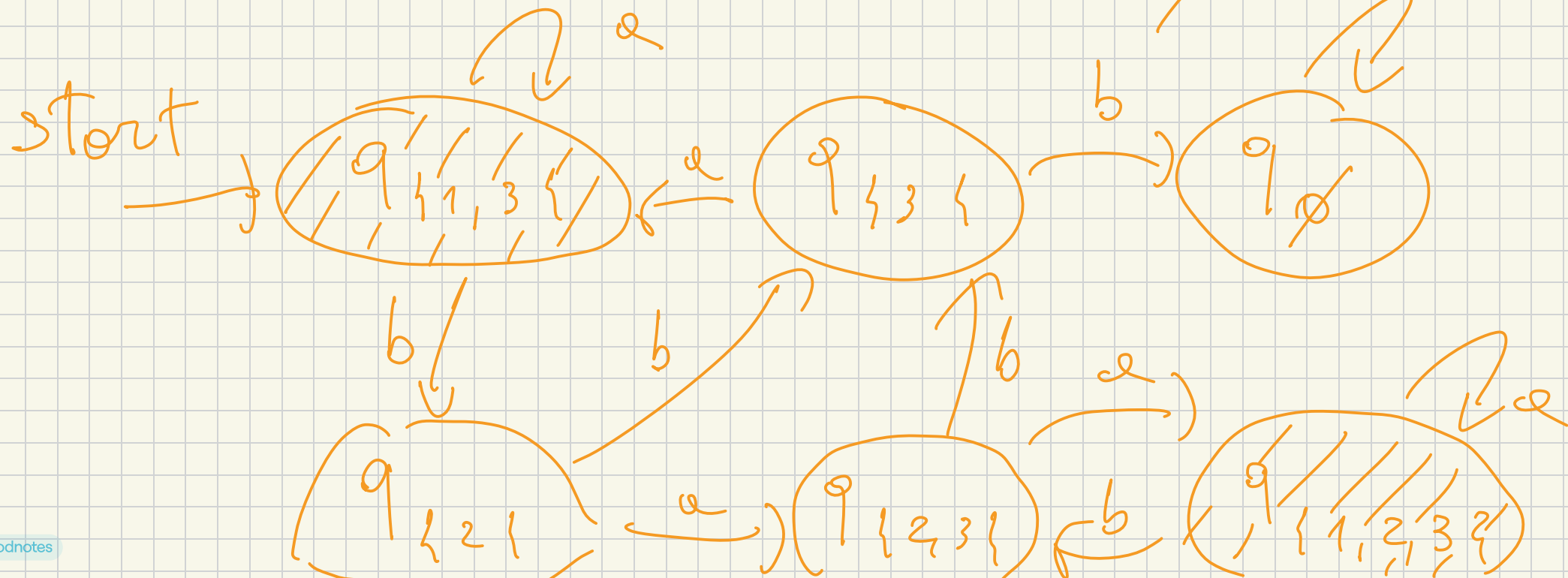
$$S_{\pi}(q_{\{1\}}, a) = q_{\emptyset}$$

$$S_{\pi}(q_{\{1\}}, b) = q_{\{2\}}$$

- Nella stato q_3 :

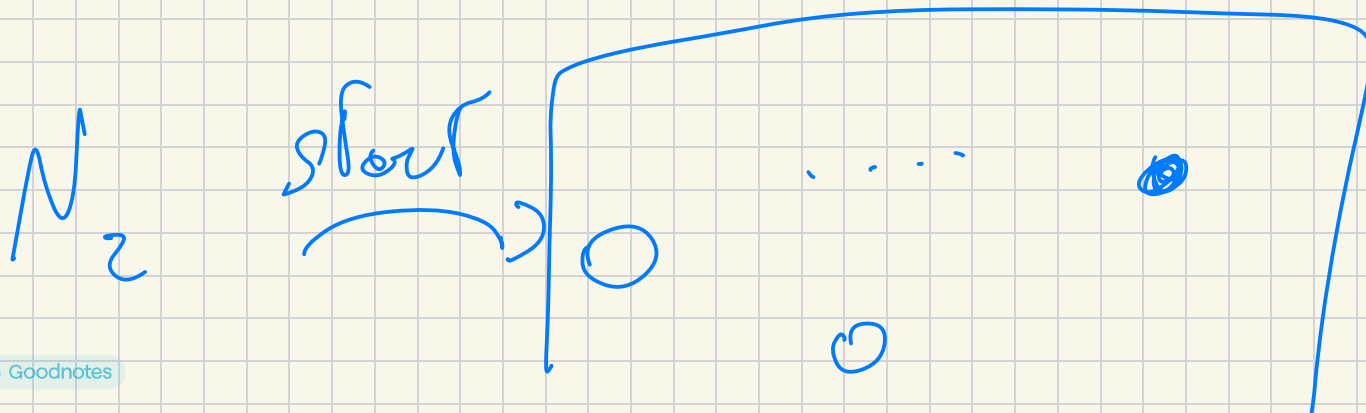
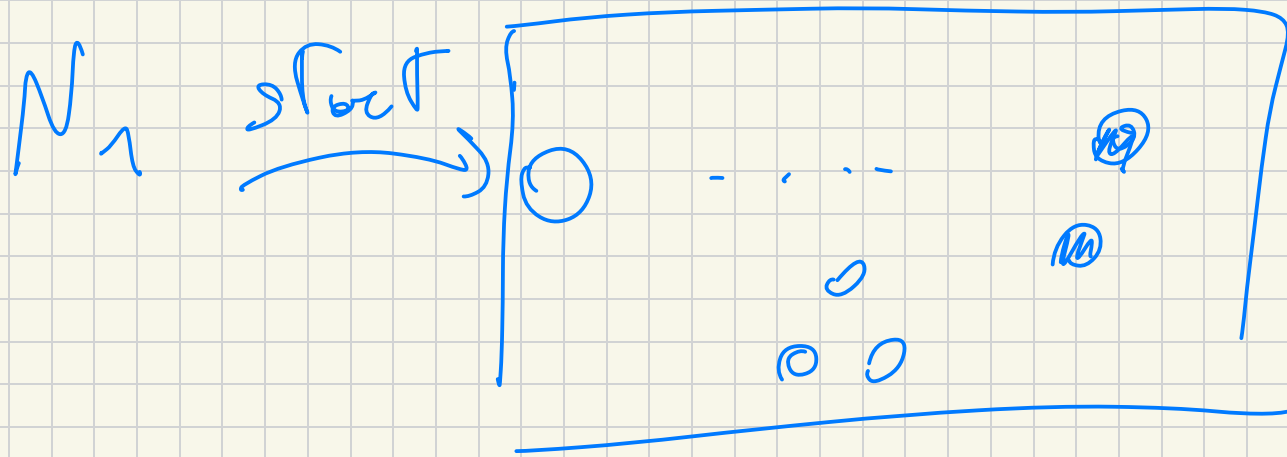
$$\delta_n(q_{\{3\}, a}) = q_{\{1, 3\}}$$

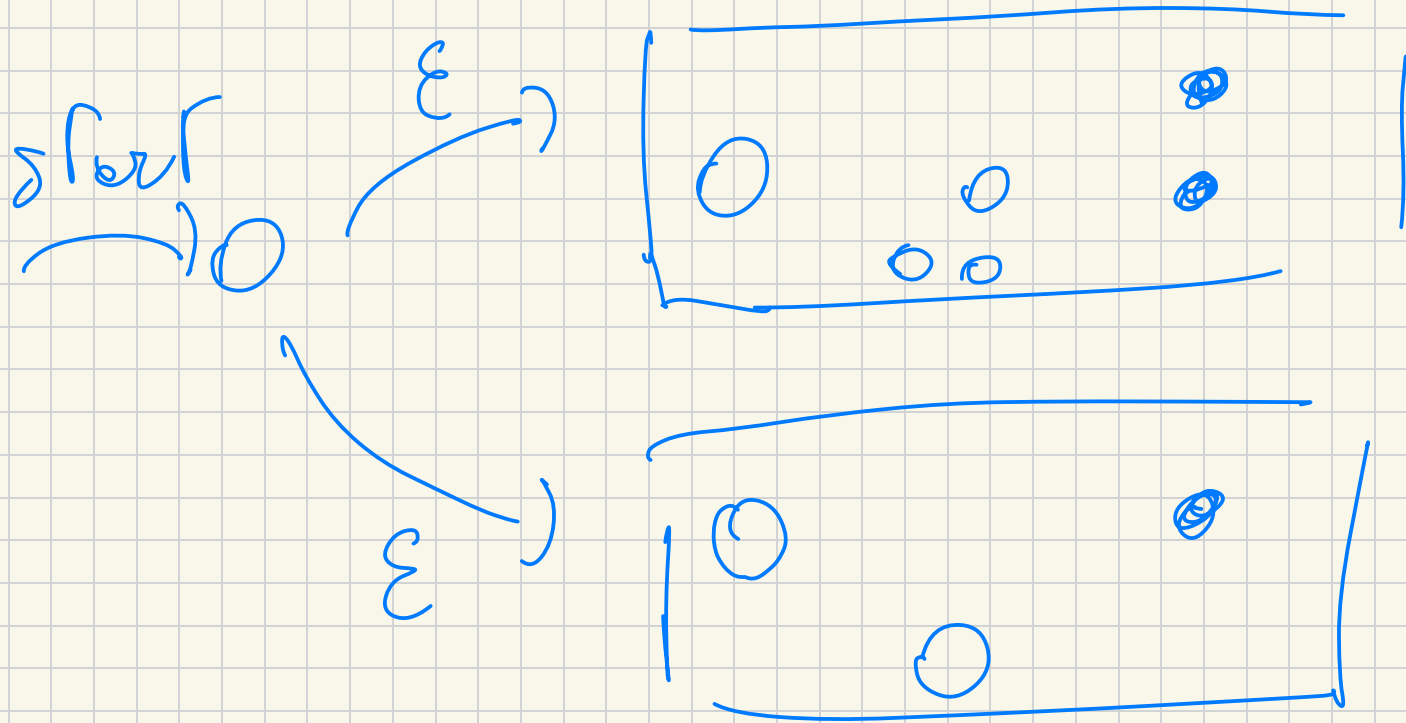
$$\delta_n(q_{\{3\}, b}) = q_{\emptyset}$$



Applichiamo il potere degli NFA. Intanto
possiamo dimostrare che RE è chiuso per $\cup$.

Dati NFA N_1, N_2 per L_1, L_2
costruiamo NFA N per $L_1 \cup L_2$

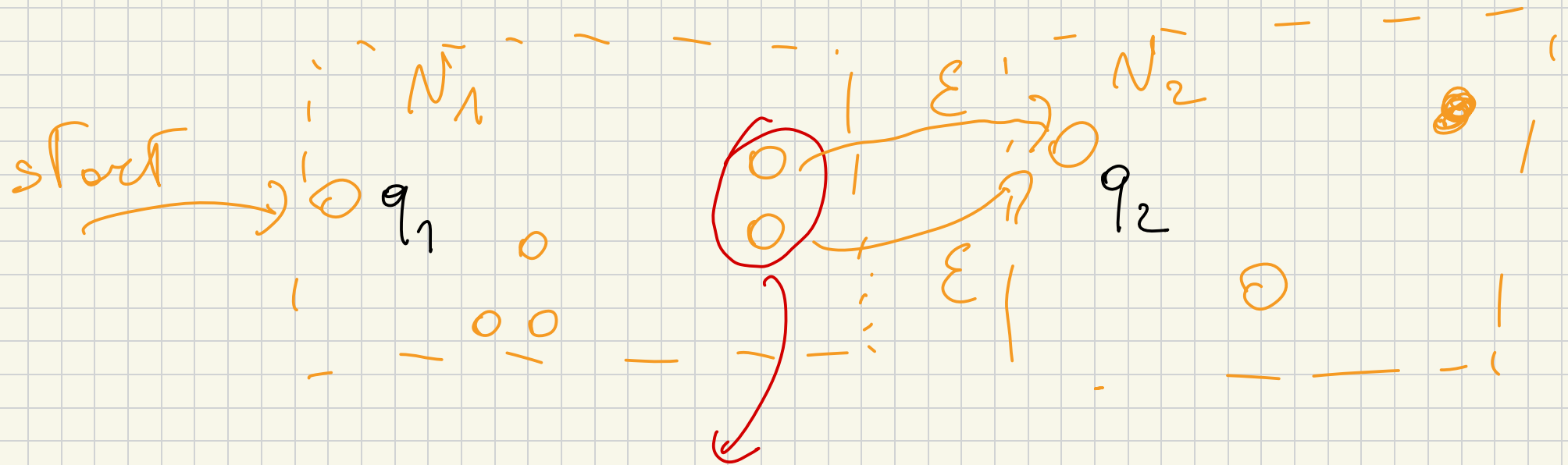




THM REG $\bar{\cup}$ CHIU SA per "o".

Defn. Defn NFA N_1, N_2 t.c., $L(N_1)$
 $= L_1$ e $L(N_2) = L_2$ costruiamo NFA
 N t.c. $L(N) = L_1 \circ L_2$

Sono N_1, N_2 come sopra.



Non sono
più FINALI

Per formulante: $N = (Q, q_0, \Sigma, \delta_N, F)$

- $q_0 = q_1$

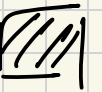
$$- Q = Q_1 \cup Q_2$$

$$- F = F_2$$

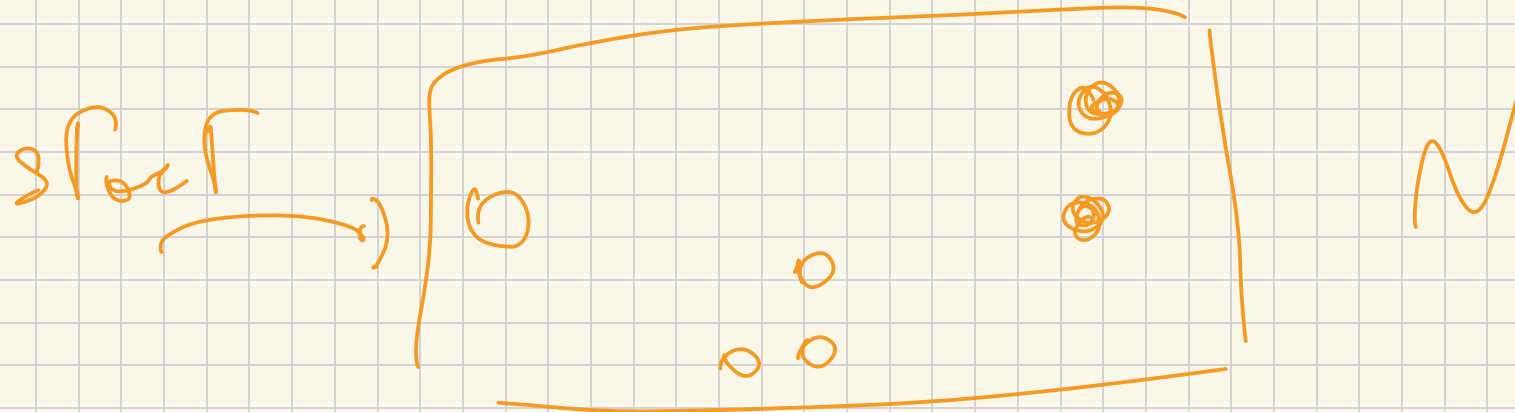
$$- \delta_N(q, a) \text{ per } q \in Q, a \in \Sigma_\varepsilon$$

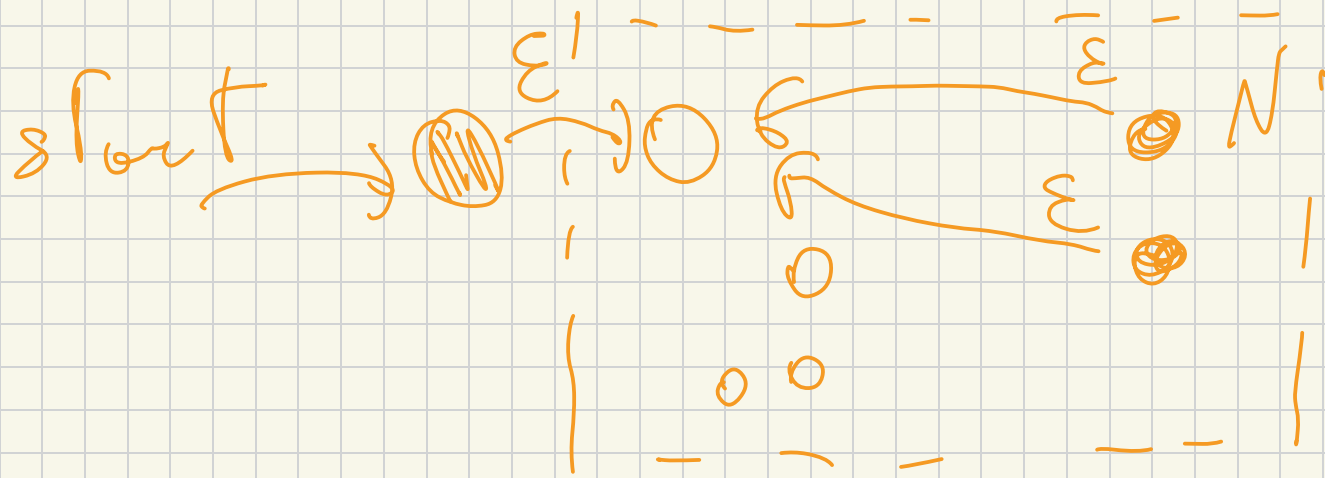
$$\delta_N(q, a) = \begin{cases} \delta_1(q, a) & q \in Q_1 \\ & q \notin F_1 \\ \delta_1(q, a) & q \in F_1 \\ & a \neq \varepsilon \\ \delta_1(q_1, a) \cup \{q_2\} & q \in F_1 \\ & a = \varepsilon \end{cases}$$

$$\left(\delta_2 (o_1, a) \quad q \in Q_2 \right)$$

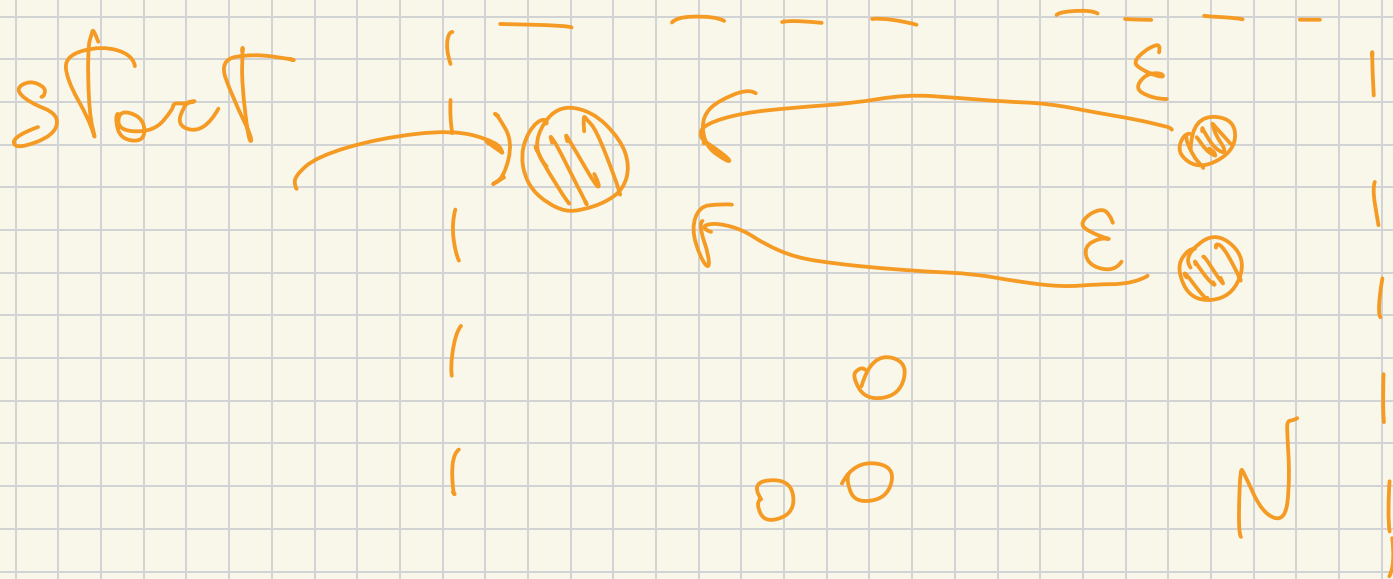


THM REG $\bar{e}$ ch+usa per $\|*\|$
 Dir. DeLo N t. c. $L(N) = L$, dove
 oppure N^* t. c. $L(N^*) = L^*$.





Esercizio: Questa è uguale ???



ESPRESSIONI REGOLARI

Strutture in logica, e per fare altre
Kleene nel 1950. Hanno ancora applicazioni nel riconoscimento
di pattern nel testo.

L'equivalente delle espressioni algebriche ma sulle stringhe

E.g. $(0 \vee 1) 0^*$ su $\Sigma = \{0, 1\}$

$$(0 \vee 1) \Rightarrow \{0\} \vee \{1\} = \{0, 1\}$$

$$0^* \Rightarrow \{0\}^*$$

$$(0 \vee 1) 0^* \Rightarrow \{0, 1\} \circ \{0\}^*$$

$$\begin{array}{l} 0^* \mid 0^* = \{ w \in \{0, 1\}^* : w \text{ contiene esattamente un } '1' \} \\ \Sigma^* \mid \Sigma^* = \text{ALMENO } // \end{array}$$

$$\downarrow$$

$$(0 \vee 1)^* \mid (0 \vee 1)^*$$

$$\Sigma^* 001 \Sigma^*$$

DEF Una ESPRESSIONE REGOLARE su Σ ($re(\Sigma)$)
 $\bar{r}$ si definisce come:

CASO BASE:

$$\left\{ \begin{array}{l} \emptyset \in re(\Sigma) \\ \varepsilon \in re(\Sigma) \\ a \in re(\Sigma), a \in \Sigma \end{array} \right.$$

CASO INDUTTIVO:

$$\left\{ \begin{array}{l} R_1 \cup R_2 \\ R_1 \circ R_2 \end{array} \right. \quad \begin{array}{l} R_1, R_2 \in re(\Sigma) \\ " \end{array}$$

(R_1^*) $R_1 \in \mathcal{R}(\Sigma)$

Alcun $\pi \in \mathcal{R}(\Sigma)$ corrisponde LINGUAGGIO $L(\pi)$.

CASO BASE

$$\left\{ \begin{array}{ll} \pi = \emptyset & L(\pi) = \emptyset \\ \pi = \varepsilon & L(\pi) = \{ \varepsilon \} \\ \pi = a & L(\pi) = \{ a \} \end{array} \right.$$

CASO INDUTTIVO:

$$\left\{ \begin{array}{ll} \pi = R_1 \cup R_2 & L(\pi) = L(R_1) \cup L(R_2) \\ \pi = R_1 \circ R_2 & L(\pi) = L(R_1) \circ L(R_2) \\ \pi = R_1^* & L(\pi) = L(R_1)^* \end{array} \right.$$

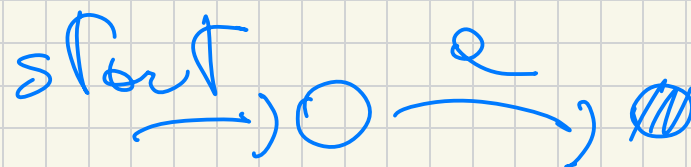
THM Un linguaggio è REGOLARE sse
esiste un'espressione regolare che lo descrive.

DIM. $\Rightarrow$ Date $\pi \in \pi(\Sigma)$, voglio
costruire DPA / NFA T.c. $L(\pi) = L(N)$
 N

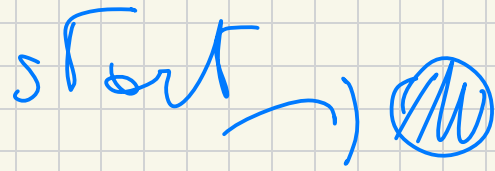
Ci viene fornito la macchina:

CASO BASE

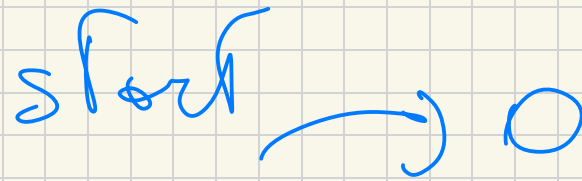
- $\pi = a \in \Sigma$



$$- \pi = \varepsilon$$



$$- \pi = \emptyset$$



Per unione se $\pi = R_1 \cup R_2, \exists$

DFA/NFA M_1, M_2 t.c. $R_1 \circ R_2$
 R_1^*

$$L(R_1) = L(M_1)$$

$$L(R_2) = L(M_2)$$

Per la chiusura
 $\text{sh } R \in R.$

$\Rightarrow \exists$ DFA/NFA M t.c. $L(M) = L(\pi)$

⇐ Def DFA / NFA M devo provare $\pi \in \pi_{\Sigma}$
t.c. $L(\pi) = L(M)$.

! Nota:

- 1) De M definiremo un equivalente GNFA (ovvero NFA generalizzato) λ cui archi hanno etichette uguali a espressioni regolari.
- 2) Vedremo algoritmo per ridurre # stati del GNFA

